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HUJI Applicative Research Report — November 2026

9 high-potential paper(s) · Monthly · November 2026 · Generated November 01, 2026

Hebrew University researchers continue to innovate across critical sectors, presenting diverse opportunities for commercial impact. This month's digest features advanced medical devices like a novel wrist arthroplasty with promising early clinical results, alongside integrated digital therapeutics poised to transform mental healthcare. Further highlights include platforms enabling next-generation drug discovery, precision oncology diagnostics, and foundational post-quantum secure communication protocols, showcasing a robust pipeline for investment and industry collaboration.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Amir Oron** — Medical Device, Clinical
- **Mor Nahum** — Medical Device, Neuroscience, Software/AI, Clinical
- **Yuval Malka** — Drug Discovery, Diagnostics, Genomics, Immunology
- **Netanel Tzarum** — Drug Discovery, Synthetic Biology, Proteomics
- **Batsheva Kerem** — Drug Discovery, Genomics, Clinical
- **Ilan Komargodski** — Software/AI, Quantum, Other
- **Nir Friedman** — Diagnostics, Genomics, Clinical
- **Mor Nitzan** — Drug Discovery, Software/AI, Genomics, Immunology
- **Emanuel Hanski** — Drug Discovery, Immunology, Proteomics, Clinical

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. First-in-human prospective pilot study of a bipolar dual-center total wrist arthroplasty: early mechanical stability and clinical outcomes. 33/50

"Novel wrist replacement shows 100% implant survivorship, offering superior biomechanics for orthopedic patients."

Main researcher: Amir Oron

Department of Orthopedic Surgery, Kaplan Medical Center, Rehovot, Israel.; Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel.

Published in Archives of orthopaedic and trauma surgery · 2026-10-22

ABOUT THIS RESEARCH

Researchers have conducted a successful first-in-human pilot study of a new bipolar dual-center wrist replacement implant designed to mimic natural wrist motion. Early results show 100% implant survivorship with no signs of loosening or dislocation after nearly two years of follow-up.

COMMERCIAL ANGLE

The development of a more biomechanically accurate wrist prosthesis addresses a significant unmet need in orthopedic surgery, offering a high-value target for MedTech venture capital and acquisition by major orthopedic device players.

COMMERCIALISATION METRICS

Novelty 8/10

The research introduces a novel bipolar dual-center design that addresses a fundamental biomechanical limitation (single center of rotation) of existing TWA implants to preserve native kinematics. As a first-in-human study of a new mechanical paradigm, it represents a significant departure from prior art in joint replacement engineering.

Commercial Potential 7/10

The device addresses a high-value unmet clinical need (implant loosening and kinematic preservation in TWA) and has successfully cleared the first-in-human milestone. However, the very small sample size (n=5) and short follow-up duration necessitate larger pivotal trials to de-risk the commercial path toward regulatory approval and market adoption.

Market Size 4/10

The target market is a niche subset of orthopedic surgery limited to end-stage wrist arthritis, which has significantly lower patient volumes compared to hip or knee arthroplasty. While clinically valuable, the addressable demand is constrained by the anatomical rarity and specific indications of the wrist joint.

Tech Readiness 7/10

The technology has successfully completed its first-in-human pilot study with positive safety and survivorship data, moving it beyond preclinical/animal models. However, it remains in the early clinical validation phase (small N=5) and requires larger, multi-center pivotal trials to establish definitive efficacy for regulatory approval.

IP Strength 7/10

The innovation introduces a novel bipolar dual-center mechanical design that addresses a known physiological limitation of existing TWA implants, suggesting strong potential for structural patentability. However, as a first-in-human pilot study with a very small sample size (n=5), the IP strength is currently limited to the device architecture rather than proven clinical utility.

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2. Home Transcranial Direct Current Stimulation With Gamified Cognitive Control Training for Depression: A Randomized Clinical Trial. 33/50

"Home-based neurostimulation with gamified cognitive training offers scalable, remote depression treatment."

Main researcher: Mor Nahum
School of Occupational Therapy, Faculty of Medicine, The Hebrew University of Jerusalem, Israel.
Published in JAMA network open · 2026 Oct 1

ABOUT THIS RESEARCH

This clinical trial investigates whether combining home-based transcranial direct current stimulation (tDCS) with gamified cognitive training is more effective for treating depression than standalone methods. The study evaluates the feasibility and efficacy of this synergistic approach in a real-world, home-use setting over a 10-week period.

COMMERCIAL ANGLE

There is a significant market opportunity for integrated digital therapeutics platforms that pair consumer-grade neurostimulation hardware with proprietary gamified software to provide scalable, remote psychiatric care.

COMMERCIALISATION METRICS

Novelty 5/10
While the combination of tDCS and gamified cognitive training addresses a known clinical gap regarding synergy, the individual components (home-based tDCS and digital therapeutics) are well-established in existing literature. The innovation lies in the specific multimodal protocol rather than a fundamental breakthrough in neurostimulation technology or a novel biological mechanism.

Commercial Potential 8/10
The study validates a synergistic combination of hardware (tDCS) and software (gamified CCT) that is highly suitable for a direct-to-consumer or prescription-based digital therapeutic model. The multicenter RCT data provides the clinical evidence required for regulatory pathways and potential licensing to established mental health platforms.

Market Size 9/10
The target market is Major Depressive Disorder (MDD), a massive global health burden with hundreds of millions of sufferers, and the home-based delivery model significantly expands the total addressable market beyond traditional clinical settings.

Tech Readiness 8/10
The technology has moved beyond the lab into a multicenter randomized clinical trial (RCT) with home-use protocols, demonstrating high clinical readiness for commercial deployment. The transition from controlled clinical settings to real-world home environments indicates the product is in the near-market stage.

IP Strength 3/10
The innovation relies on combining existing, well-known technologies (tDCS and cognitive training) rather than a novel hardware or neurostimulation method, making it difficult to secure broad, defensive patent protection. The 'gamified' element and procedural synergy are likely considered incremental improvements or methods of use, which offer limited long-term defensibility against competitors.

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3. RNA dicing promotes the expression of an oncogenic JAK1 isoform. 33/50

"New 'RNA dicing' discovery enables precision cancer therapy by stratifying patients for existing drugs."

Main researcher: Yuval Malka · y.malka@mail.huji.ac.il.

Department of Biochemistry and Molecular Biology, The Institute for Medical Research Israel-Canada, Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem 9112001, Israel. Electronic address: y.malka@mail.huji.ac.il.

Published in Cell reports · 2026-04

ABOUT THIS RESEARCH

Researchers discovered a process called 'RNA dicing' where mRNA is cleaved to produce a truncated, oncogenic version of the JAK1 protein. This process changes how cells grow and contributes to cancer development, particularly in endometrial tumors.

COMMERCIAL ANGLE

The discovery enables a new precision medicine approach by using RNA dicing profiles to stratify patients for existing JAK1 inhibitors like momelotinib.

COMMERCIALISATION METRICS

Novelty 9/10

The discovery of 'RNA dicing' represents a paradigm-shifting mechanism that challenges the traditional ORF-centric view of translation, offering a novel biological principle. It provides a highly innovative foundation for new therapeutic strategies and patient stratification models based on non-canonical mRNA processing.

Commercial Potential 7/10

The discovery provides a high-value biomarker for patient stratification in oncology, specifically linking 'RNA dicing' to drug sensitivity for existing JAK inhibitors like momelotinib. While the mechanistic insight is profound, the commercial leap depends on validating this as a diagnostic tool or developing new therapeutics targeting the diced isoform specifically.

Market Size 7/10

The discovery identifies a novel mechanism of protein expression that serves as a high-value biomarker for patient stratification in oncology, specifically for existing JAK inhibitors. While the mechanism itself is a potentially platform-enabling biological insight, the immediate market is constrained to specific cancer indications and the repurposing/refinement of existing drug classes.

Tech Readiness 3/10

This is fundamental mechanistic research identifying a new biological phenomenon (RNA dicing) and its role in oncogenesis. While it suggests potential for patient stratification using existing inhibitors, it remains at the basic discovery/proof-of-concept stage without a developed diagnostic or therapeutic pipeline.

IP Strength 7/10

The discovery of 'RNA dicing' as a mechanism for generating oncogenic isoforms provides a strong foundation for new therapeutic targets and diagnostic biomarkers. While the biological mechanism is novel and potentially patentable, the IP strength is tempered by the fact that the study focuses on a specific endogenous process rather than a proprietary platform technology.

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4. Modular Platform for Recombinant Production of Highly Toxic Bacterial Proteins. 32/50

"Modular platform enables large-scale production of highly toxic proteins for next-gen antimicrobials."

Main researcher: Netanel Tzarum

Department of Biological Chemistry, Alexander Silberman Institute of Life Sciences, Faculty of Mathematics & Science, The Hebrew University of Jerusalem, Jerusalem, Israel.; The Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Jerusalem, Israel.

Published in Microbial biotechnology · 2026 Oct 17

ABOUT THIS RESEARCH

Researchers developed a modular system to produce highly toxic bacterial proteins by fusing them to neutralizing immunity proteins to prevent host cell death. They further optimized the process using intracellular proteases and computational redesign to increase yields and simplify purification.

COMMERCIAL ANGLE

Enables the large-scale production and screening of previously 'undruggable' bacterial toxins for use as next-generation antimicrobial agents or precision therapeutics.

COMMERCIALISATION METRICS

Novelty 7/10

The work provides a modular, platform-enabling approach to unlock previously refractory toxic proteins, though the core concept of toxin-immunity fusion is an existing biological principle. The novelty lies in the streamlined engineering workflow and the use of computational redesign to optimize these fusions for industrial-scale recombinant production.

Commercial Potential 7/10

The modular platform addresses a clear bottleneck in drug discovery and biotechnology by unlocking access to previously unproducible high-value molecules. While it is a highly scalable platform-enabling technology, its commercial value is contingent on the downstream utility of the specific toxins discovered via this method.

Market Size 7/10

The platform enables access to a vast, untapped library of bioactive toxins for antimicrobial and biotech discovery, representing a high-value upstream enabler for drug discovery. However, as an upstream production tool, its market size is constrained by the size of the R&D service and reagent markets rather than direct therapeutic sales.

Tech Readiness 4/10

The technology is at the proof-of-concept stage, demonstrating improved yields and workflow streamlining in a laboratory setting for specific protein families. While it establishes a functional modular platform, it lacks the scale-up validation, industrial process optimization, or application-specific testing required for higher TRL scores.

IP Strength 7/10

The modular fusion-and-cleavage platform represents a highly defensible, platform-enabling technology with clear patentability for the specific fusion architectures and protease-mediated release mechanisms. While technically robust, the score is capped slightly below an 8 as the core concept of toxin-immunity pairing is a biological principle that may face challenges in claiming broad exclusivity against evolving computational redesign methods.

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5. SPL84 and trikafta® exhibit comparable and additive effects in patient-derived HBE cells carrying the 3849+10kb C→T/F508del CFTR variants. 31/50

"Combination therapy significantly improves CFTR function for underserved cystic fibrosis patients."

Main researcher: Batsheva Kerem · batshevak@savion.huji.ac.il.

SpliSense, Haddasah Ein Kerem, Jerusalem, Israel; Department of Genetics, The Hebrew University, Jerusalem, Israel. Electronic address: batshevak@savion.huji.ac.il.

Published in Journal of cystic fibrosis : official journal of the European Cystic Fibrosis Society · 2026-06

ABOUT THIS RESEARCH

Researchers found that combining a new inhaled antisense oligonucleotide (SPL84) with the existing drug Trikafta® provides additive benefits for patients with a specific cystic fibrosis mutation. This combination therapy effectively restores CFTR protein function in human lung cells, potentially offering better clinical outcomes than current standards of care alone.

COMMERCIAL ANGLE

Developing a combination therapy or a precision medicine regimen that pairs existing CFTR modulators with targeted antisense oligonucleotides to capture the underserved segment of patients with modest responses to current treatments.

COMMERCIALISATION METRICS

Novelty 6/10

The research is an incremental optimization study focused on combination therapy rather than a fundamental discovery of a new biological mechanism or platform. While the additive effect of an ASO with an existing standard of care is clinically relevant, it represents an evolutionary application of known splicing modulation and CFTR corrector technologies.

Commercial Potential 8/10

The study demonstrates a clear additive therapeutic strategy for a specific, high-value patient population, providing a strong de-risking signal for a combination product approach. The focus on an inhaled ASO targeting a known clinical gap in current standard-of-care (Trikafta) represents a highly viable and direct path to clinical application.

Market Size 4/10

The target market is limited to a highly specific genetic sub-population (3849+10kb variant) within the already established cystic fibrosis patient base. While clinically valuable for non-responders, the addressable demand is niche and constrained by the prevalence of this specific splicing mutation.

Tech Readiness 6/10

The asset has already completed a Phase 2a clinical trial, indicating significant de-risking; however, the current paper focuses on ex vivo validation in patient-derived cells rather than advancing the clinical trial phase.

IP Strength 7/10

The innovation utilizes a specific antisense oligonucleotide (ASO) sequence targeting a known splicing defect, which is highly patentable and provides a clear pathway for composition-of-matter protection. However, the defensibility is slightly tempered by the fact that it is a combination therapy with an existing standard-of-care, which may face competition from other splicing modulators.

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6. Scalable distributed agreement from LWE: Byzantine agreement, broadcast, and leader election. 30/50

"New LWE-based protocol provides scalable, post-quantum secure consensus for decentralized networks."

Main researcher: Ilan Komargodski

School of Computer Science and Engineering, The Hebrew University of Jerusalem, Givat Ram, Jerusalem, Israel.; NTT Research, Sunnyvale, CA USA.

Published · 2026 Oct 1

ABOUT THIS RESEARCH

This paper proposes a new method for achieving consensus in large networks of computers using Learning With Errors (LWE) cryptography. It allows many different nodes to agree on information efficiently even if some of those nodes are malicious or failing.

COMMERCIAL ANGLE

The research provides a foundation for highly scalable, post-quantum secure communication protocols essential for decentralized data infrastructure and secure multi-party computation.

COMMERCIALISATION METRICS

Novelty 8/10

The paper introduces a significant paradigm shift by replacing standard cryptographic assumptions with Learning With Errors (LWE) to achieve sublinear communication complexity, offering a potentially quantum-resistant path to scalability in distributed consensus. It directly improves upon the state-of-the-art by bridging the gap between computational efficiency and post-quantum security assumptions.

Commercial Potential 7/10

The work addresses a critical bottleneck in scalable blockchain and decentralized infrastructure by leveraging LWE-based cryptography for efficient consensus. While it offers a highly licensable foundational layer for next-generation distributed systems, its commercial value is currently tethered to the broader adoption of post-quantum cryptographic standards in large-scale networks.

Market Size 9/10

Consensus and Byzantine agreement protocols are foundational infrastructure for the entire blockchain and distributed systems market, representing a multi-billion dollar addressable demand. By solving for sublinear communication complexity, this research directly addresses the critical scaling bottleneck that currently limits the global adoption of decentralized financial and enterprise-grade distributed ledgers.

Tech Readiness 3/10

The paper focuses on fundamental cryptographic theory and complexity bounds (LWE-based protocols) rather than a practical implementation or system architecture. It represents early-stage mathematical research that provides the building blocks for future scalable distributed systems but lacks the engineering maturity for immediate commercial application.

IP Strength 3/10

This is a fundamental theoretical contribution in cryptographic protocol design, which typically lacks the specific embodiment or novel physical/methodological utility required for robust, platform-enabling patent protection. While highly impactful for the field of distributed systems, it constitutes mathematical methodology and algorithmic theory that is broadly challenging to ring-fence as proprietary intellectual property.

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7. Plasma nucleosome profiling reports on tumor burden and molecular subtypes in small cell lung cancer. 28/50

"Novel liquid biopsy tracks tumor burden and subtypes in lung cancer, enabling personalized treatment."

Main researcher: Nir Friedman

School of Computer Science and Engineering, Hebrew University of Jerusalem, Jerusalem, Israel.; Lautenberg Center for Immunology and Cancer, Institute for Medical Research Israel-Canada, The Hebrew University-Hadassah Medical School, Jerusalem, Israel.; Hebrew University Center for Computational Medicine, The Hebrew University-Hadassah Medical School, Jerusalem, Israel.

Published in The Journal of clinical investigation · 2026 Oct 1

ABOUT THIS RESEARCH

Researchers have developed a method to analyze nucleosomes in blood plasma to monitor tumor burden and identify molecular subtypes in small cell lung cancer patients. This liquid biopsy approach provides a non-invasive way to track disease progression and characterize cancer genetics through circulating cell-free DNA structures.

COMMERCIAL ANGLE

There is significant commercial potential in developing a high-sensitivity liquid biopsy diagnostic platform for real-time oncology monitoring and personalized treatment selection.

COMMERCIALISATION METRICS

Novelty 5/10

While liquid biopsy applications are high-value, nucleosome profiling is a known methodology in the burgeoning field of cell-free DNA fragmentomics, making this specific application to SCLC an incremental advancement rather than a fundamental breakthrough.

Commercial Potential 8/10

The technology targets high-value clinical unmet needs in oncology, specifically tumor burden and molecular subtyping, which are highly licensable or developable as liquid biopsy diagnostic assays. Profiling nucleosomes in plasma represents a scalable, platform-enabling approach for precision medicine in lung cancer.

Market Size 7/10

Small cell lung cancer represents a high-unmet-need oncology segment, but the specific application to nucleosome profiling for tumor burden may be limited to a niche subset of liquid biopsy monitoring rather than broad-spectrum screening.

Tech Readiness 4/10

The study focuses on profiling nucleosomes in plasma, which represents a liquid biopsy research phase characterizing biomarkers rather than a validated clinical diagnostic tool. While it demonstrates proof-of-concept for tumor burden monitoring, it remains in the analytical validation/discovery stage typical of early-to-mid-stage life science research.

IP Strength 4/10

The innovation appears to be a diagnostic application of existing biological markers (nucleosome profiling) rather than the creation of a novel, patentable molecule or device. Defensibility is likely limited to specific proprietary algorithms or data sets, which are harder to protect than foundational biotechnology.

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8. Identifying maximally informative signal-aware representations of single-cell data using the information bottleneck. 28/50

"AI framework distills complex single-cell data, accelerating drug discovery and precision medicine."

Main researcher: Mor Nitzan · mor.nitzan@mail.huji.ac.il.

School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel; Racah Institute of Physics, The Hebrew University of Jerusalem, Jerusalem, Israel; Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. Electronic address: mor.nitzan@mail.huji.ac.il.

Published in Cell systems · 2026-05

ABOUT THIS RESEARCH

This paper introduces bioIB, an AI framework that uses the information bottleneck method to distill complex single-cell RNA sequencing data into highly specific, interpretable gene clusters. By filtering out noise and focusing on relevant biological signals like disease states, it maps out the hierarchical relationships between different cell types and biological processes.

COMMERCIAL ANGLE

The framework can be commercialized as a high-throughput software platform for drug discovery and precision medicine to identify specific cellular biomarkers and disease drivers from noisy single-cell datasets.

COMMERCIALISATION METRICS

Novelty 6/10

The paper applies an established information theory framework (Information Bottleneck) to a well-trodden domain (scRNA-seq), representing an incremental methodological improvement rather than a fundamental scientific breakthrough. While the application to gene clustering and hierarchical structure is useful, it follows existing trends in deep learning-based dimensionality reduction and feature selection.

Commercial Potential 6/10

The framework presents a strong platform-enabling computational tool for drug discovery and biomarker identification, but its commercial value is currently tied to software/algorithm licensing rather than a standalone proprietary product. Its utility depends on integration into existing bioinformatics pipelines or single-cell analysis platforms.

Market Size 8/10

The framework acts as a platform-enabling technology for the massive and growing single-cell multi-omics market, offering high value for drug discovery and precision medicine. By optimizing signal extraction in complex datasets, it addresses a critical bottleneck in the multi-billion dollar bioinformatics and clinical diagnostics sectors.

Tech Readiness 4/10

The paper presents a computational framework and algorithmic methodology (bioIB) rather than a physical product or clinical tool. While it demonstrates utility on existing datasets, it remains at the fundamental software/algorithm development stage (TRL 3-4) rather than a validated industrial platform.

IP Strength 4/10

The innovation is primarily a computational framework based on the Information Bottleneck method, which is an established theoretical principle in machine learning, making the core algorithm difficult to patent. While the application to scRNA-seq is valuable, the defensibility is low as it likely falls under 'mathematical methods' and is susceptible to rapid replication by open-source community developments.

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9. Group A Streptococcus host-pathogen dual crosstalk. 28/50

"Discovered bacterial metabolic vulnerability offers a novel target for anti-infective drug development."

Main researcher: Emanuel Hanski · emanuelh@ekmd.huji.ac.il.
Department of Microbiology and Molecular Genetics, The Institute for Medical Research, Israel-Canada (IMRIC), Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem 9112102, Israel. Electronic address: emanuelh@ekmd.huji.ac.il.
Published in Trends in microbiology · 2026-06

ABOUT THIS RESEARCH

Researchers have discovered that Group A Streptococcus bacteria hijack the human cell's stress response to force the production of the amino acid asparagine, which the bacteria then consume to fuel infection. This mechanism creates a metabolic bridge between host cell stress and bacterial virulence, mimicking similar pathways seen in cancer cells.

COMMERCIAL ANGLE

The discovery of the PERK-eIF2α-ATF4-Asn axis provides a novel metabolic target for developing small-molecule inhibitors to starve bacterial pathogens or disrupt their ability to exploit host resources.

COMMERCIALISATION METRICS

Novelty 8/10
The research identifies a highly specific, previously unknown metabolic hijacking mechanism where an extracellular bacterium exploits a host stress pathway (PERK-eIF2α-ATF4) to drive nutrient biosynthesis. This discovery of a direct 'crosstalk' circuit provides a novel, targetable metabolic vulnerability that parallels oncogenic pathways, offering significant potential for new antimicrobial strategies.

Commercial Potential 6/10
The identification of a specific metabolic axis (ATF4-Asn) provides a clear target for novel antimicrobial drug development, though the transition from mechanistic discovery to a viable therapeutic lead remains speculative.

Market Size 7/10
The research targets Group A Streptococcus, a major global pathogen with high clinical burden, but its most significant market value lies in the potential to translate the ATF4-Asn metabolic axis into oncology therapeutics. While the infectious disease application is substantial, the broader addressable market is expanded by the cross-sector relevance to cancer metabolism and therapy resistance.

Tech Readiness 2/10
The paper describes fundamental mechanistic biology and discovery of a signaling pathway in a laboratory setting. It lacks any mention of therapeutic intervention, drug development, or preclinical validation required to move beyond basic research.

IP Strength 5/10
The paper describes a fundamental biological mechanism (the ATF4-Asn metabolic axis) rather than a specific therapeutic molecule or engineered platform. While the discovery provides a targetable pathway, the core finding is a natural phenomenon which is difficult to patent without specific, non-obvious chemical or biological interventions.

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